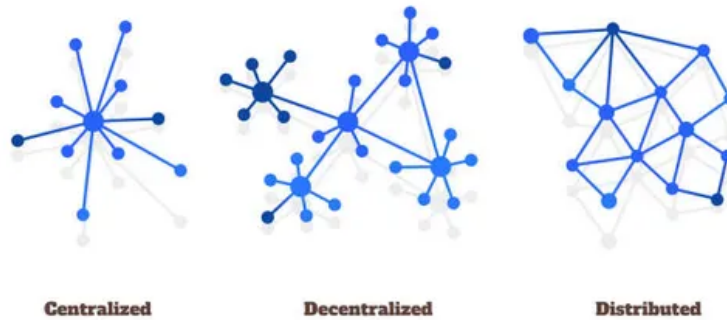


Why "Decentralized Systems" and are they reliable?



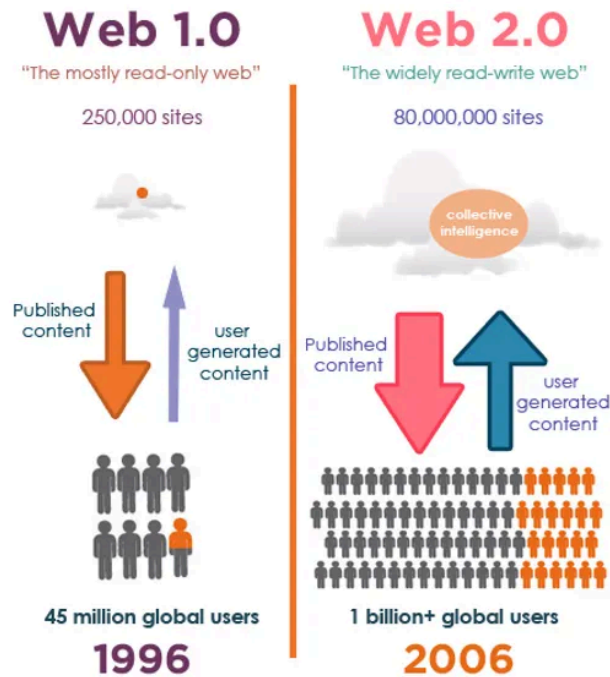
Centralized Systems - An Analogy

Imagine you're leading a club at your college or a lead initiative at your workplace. You're not only leading the activities, you are simultaneously managing and getting them all done by yourself. Your team is only there to fulfill an eligibility/criteria. You're the centralized piece of this web. Let's say you take an emergency off-day and the particular day happens to be an important day for a club/initiative showcase. What happens? The entire facade collapses. No one else is able to substitute for you because they were never actively involved in the activities. They may try and prepare at the last minute but they're ultimately bound to fumble on stage revealing the fact that you were the only person carrying the whole team. Can we not all relate to this to some extent?

In this case, centralization was also detrimental to the main component, AKA you. You were exhausted and burnt out from the whole "I'll do everything" ordeal. In the context of Web 2.0 however, as we'll see in the next section, centralization actively benefits the key entities that are involved. Ever heard of a "monopoly"?

Centralized Systems - A Formal Definition

Before we move on to decentralized systems, it's most important that we wrap our heads around Web 2's (Dynamic web or most of present-day web) centralized systems, first.



Centralization in Web 2.0 is evident in the dominance of a few large technological firms taking a majority control of user data, identities and platforms. Some well-known examples include Meta, Snapchat, Amazon, YouTube and Spotify. Almost everyone knows these application names. They're used by most people that we know. If someone drops a new social app's name, you're like "Who?" because it is simply not popular. All this user attention is heavily centralized towards a few organizations that monopolize and profit from our data.

Why Decentralized Systems?

Decentralized systems, in terms of blockchain and crypto, refer to the concept of power being distributed across multiple nodes/persons instead of lying with a central, authoritative entity.

We'll go back to the team example for this one. If other team members also actively participated in driving the initiative to success instead of pushing the team leader to get it all done themselves, they may be able to avoid disaster when the leader is absent during a critical evaluation period.



Now, coming back to web2, the original internet, as envisioned in the 1990s, was built on open, distributed protocols (a set of rules that govern how something like messaging, file transfer or email transfer must be executed) like HTTP, DNS, FTP, and SMTP. These protocols enabled equal access and individual or collective innovation without reliance on any single/centralized corporate entity. This decentralized architecture was designed to be democratic and open to all.

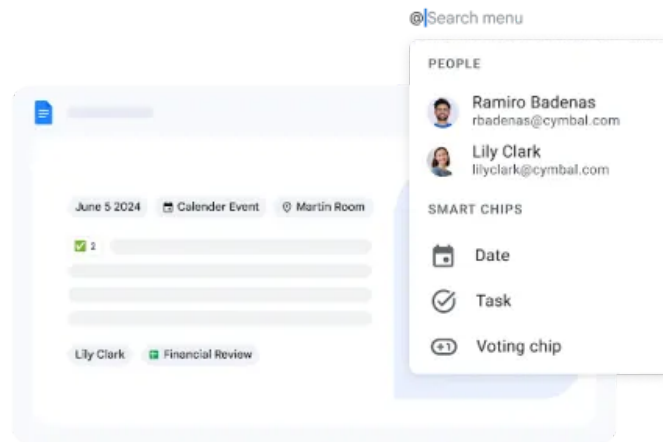
By including multiple nodes and ensuring that authority is spread among many, it provides trust within the ecosystem. A big benefit is users being able to partake in peer-to-peer transactions with more peace of mind without having to rely on intermediaries. How? Because transactions are publicly visible to the network participants and due to smart contracts, which are, in short, digital programs stored on blockchain networks, they are also immutable.

Similar to how a team can catastrophically collapse when the centralized member is absent, centralized systems also pose a severe risk of failure relative to decentralized systems as a single point of critical failure can affect billions of systems around the world. Aside from security benefits and distribution of power, a blockchain network is decentralized to also eliminate any single point of failure. If one node on the chain fails, the other network nodes will continue to operate to bolster resilience.

How Decentralized Systems Actually Work

So we know why decentralization matters, but how does it actually function without a central authority calling the shots?

Think of a decentralized network as a shared Google Doc that thousands of people have a copy of, except no one person owns it, no one can secretly edit it, and every change made is visible to everyone. This shared record is what we call a blockchain, a continuously growing chain of "blocks," each containing a set of verified transactions or data.

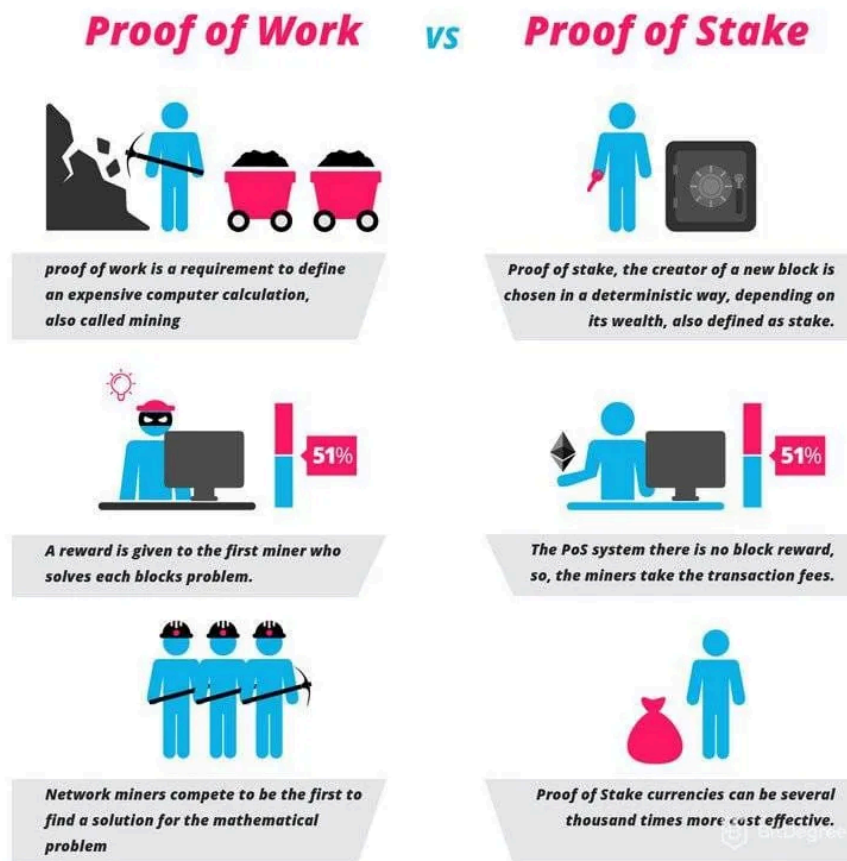


When you initiate a transaction on a blockchain network, it doesn't just get processed by a single server somewhere. Instead, it gets broadcast to a network of nodes, essentially computers participating in the network, that collectively verify whether the transaction is valid. Once verified, it gets bundled into a block and added to the chain permanently.

But how do thousands of nodes, with no central coordinator, agree on what's valid? That's where consensus mechanisms come in. The two most well-known are:

Proof of Work (PoW): Nodes (called miners) compete to solve complex mathematical puzzles. The first to solve it gets to add the next block and is rewarded with cryptocurrency. Bitcoin uses this method.

Proof of Stake (PoS): Instead of computational competition, validators are chosen based on how much cryptocurrency they "stake" or lock up as collateral. Ethereum transitioned to this model as it is far more energy-efficient.



Both mechanisms ensure that tampering with the blockchain is extraordinarily difficult. To alter a single block, you'd have to redo the verification for every block that came after it, across the majority of nodes simultaneously. In practice, this makes fraud nearly impossible.

The Limitations and Challenges of Decentralized Systems

Decentralization sounds almost utopian on paper. No single point of failure, no monopolistic control, peer-to-peer trust. So why hasn't it completely replaced the centralized systems we use today? It comes with real trade-offs.

The most well-known challenge is called the Blockchain Trilemma, a term coined by Ethereum co-founder Vitalik Buterin. It suggests that a blockchain can realistically only optimize for two out of three properties at any given time: decentralization, security, and scalability. If you prioritize decentralization and security, you likely sacrifice speed and the ability to handle large transaction volumes. It's a constant balancing act that blockchain developers continue to deal with.



Then there's the environmental concern. Proof of Work consensus, as used by Bitcoin, demands enormous amounts of computational power and therefore electricity. At its peak, Bitcoin's annual energy consumption has been compared to that of entire countries. While Proof of Stake has emerged as a greener alternative, the conversation around crypto's carbon footprint is far from over.

There's also a very human problem: accessibility and usability. The average person finds blockchain technology intimidating. Managing private keys, understanding wallet addresses, navigating gas fees, none of it is intuitive. Until decentralized systems become as seamless to use as opening an Instagram account, mainstream adoption will remain an uphill battle.

Finally, there's a certain irony worth acknowledging. Despite the decentralization ideal, many platforms built on blockchain technology have gradually become centralized in practice. A handful of large mining pools, venture-backed exchanges, and dominant validators wield disproportionate influence. Whether true decentralization is achievable at scale, or simply an ideal to strive toward, is a question the space is still working through.